



Power BI

MPBI01 – EXERCISES (ENG)



Power BI

Desktop

Ludovic Moreau
moreau@artofnet.ch

Queries [8]

- Medal Sort table
- Paris2024_ATHLETES
- Paris2024_MEDALS
- Paris2024_NOCS
- Paris2024_EVENTS
- dimDisciplines
- 01.Measures
- Z IMG

fx = Table.RenameColumns(#"Paris2024_ATHLETES développé",{"code", "AthleteID"})

medal_code	medal_date	A ^B name	1 ² NOCID	A ^B compound key	1 ² DisciplineID	A ^B Name in Athletes table	1 ² AthletelD
1	07.08.2024	Artur ALEKSANYAN	10	artur aleksanyan - 10		316 ALEKSANYAN Artur	1532872
2	07.08.2024	Malkhas AMOYAN	10	malkhas amoyan - 10		314 AMOYAN Malkhas	1532873
3	11.08.2024	Tatiana RENTERIA REN...	48	tatiana renteria renteria...		323 RENTERIA RENTERIA Tatia...	1535513
4	09.08.2024	Shanieka RICKETTS	106	shanieka ricketts - 106		59 RICKETTS Shanieka	1536888
5	26.07.2024	Federico Nilo MALDINI	104	federico nilo maldini - 1...		217 MALDINI Federico Nilo	1551386
6	28.07.2024	Paolo MONNA	104	paolo monna - 104		217 MONNA Paolo	1551387
7	09.08.2024	Marco Alonso VERDE A...	133	marco alonso verde alvare...		84 VERDE ALVAREZ Marco Alo...	1535187
8	30.07.2024	Daniel WIFFEN	99	daniel wiffen - 99		261 WIFFEN Daniel	1539969
9	04.08.2024	Daniel WIFFEN	99	daniel wiffen - 99		262 WIFFEN Daniel	1539969
10	08.08.2024	Osmar OLVERA IBARRA	133	osmar olvera ibarra - 133		138 OLVERA IBARRA Osmar	1535345

Query Settings

PROPERTIES

Name

Paris2024_MEDALS

All Properties

APPLIED STEPS

Source

En-têtes promus

Type modifié

Autres colonnes supprimées

Colonnes permutées

CREATING THE DATA MODEL



Power BI
Desktop



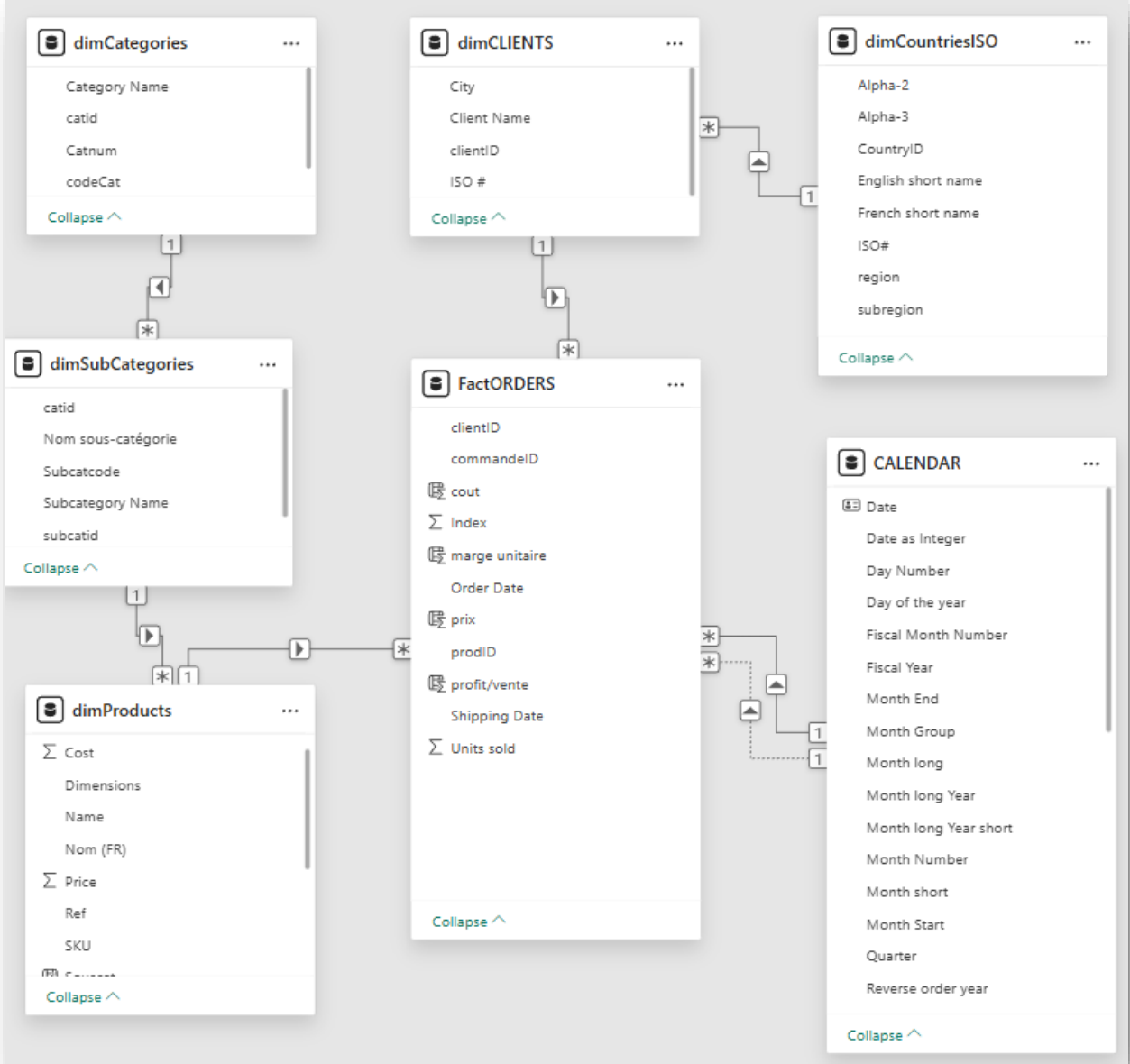
1. Combine all rows (766) of the files contained in the **Commandes-Orders folder** into a FactOrders query.
2. The Swicrosoft_prod_prix file contains both tables tbl_prodcats and tbl_prod. Merge them with the [Number] field. Rename to dimProducts. Remove the option to load the 2 original tables so as not to load them into the data model and move them to a 'Noload' group (to be created in the Power Query panel).
3. Create a new Access query and select the **SwiCrosoft_Données.accdb** database. Select and load the two Categories and Subcategories tables. Rename to dimCat and dimSubcat
4. Add the customers table from the CSV file **SwiCrosoft_CLIENTS_master.csv**. Rename to dimClients. CAUTION: Select the semicolon in the column separator and click on the 'Advanced' button to select UTF-8 in the encoding if not detected by Power BI.
5. Add the dimCountriesISO table from the PDF file **CountriesISO3166.pdf**. Choose a single page from the list and choose 'transform' then filter the list of the 9 tables included in the document and extract the Data column then promote the headers and detect the type.
6. Add a query on the JSON file **'CountriesContinentsJSON.json'**. Convert to table and expand column 1 keeping only [name],[cca2], [cca3], [region], and [subregion]. Expand [name] with only [common]. Disable loading and move to the 'Noload' group.
7. Merge dimCountriesISO with CountriesContinentsJSON and add [region] and [subregion] to the country table. Detect types and then exit Power Query (Home> Close & Apply).

Generate the model view

Navigate to the Model view mode in Power BI

1. In the model view, create the necessary relationships between tables.
2. Add a Calendar table by inserting the DAX script provided for this purpose. Choose the start and end year.
Mark as date table and select the date field in the calendar table as the date reference.
3. Create a manual table (Home > Enter Data) for storing measures. Rename '01 Measures'.
4. Create a 'hr_cat' hierarchy in the dimProducts table and add [Cat], [Subcat] and [Ref].
5. Create a 'hr_year' hierarchy in the Calendar table and add [Year], [Quarter], [Month Short] and [Date].

Multi-source data model



Create calculated columns

1. Add two columns to display the cats and subcats in the dimProducts table:

Cat = `RELATED(dimCat[Category Name])`

Subcat = `RELATED(dimSubCat[Subcategory Name])`

2. Hide the cat and subcat tables from the diagram mode.

3. Add two calculated columns to the factOrders table that evaluates the total per order line and then the profit per sale:

Total/Sale = `RELATED(dimProducts[Price(CHF)]) * factOrders[NB Sale]`

Profit/Sale = `(RELATED(dimProducts[Price(CHF)]) - RELATED(dimProducts[Cost(CHF)])) * factOrders[NB Sale]`

Create a Measure Table

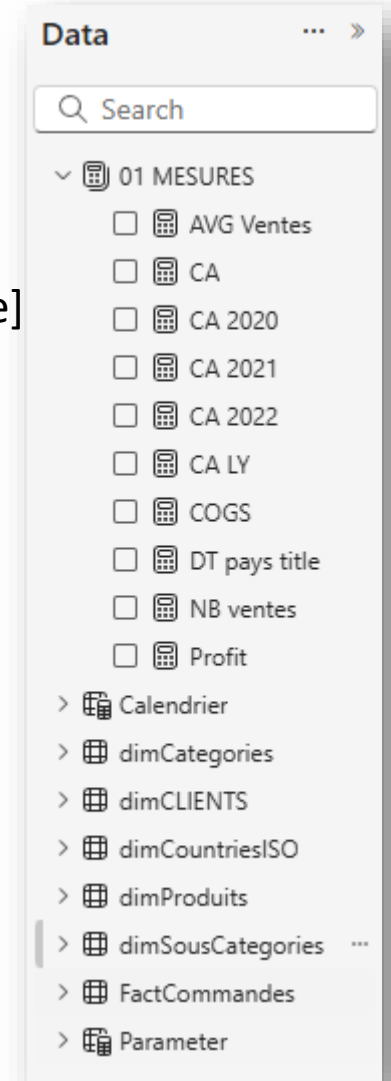
1. Add the following measures:

Revenue = **SUMX**(FactCommandes,FactCommandes[NB vente]***RELATED**(dimProduits[Price])

Profit = **SUM**(FactVentes[Profit])

COGS = **SUMX**(FactCommandes,FactCommandes[NB vente]*FactCommandes[cost])

NB sales = **COUNTROWS**(FactCommandes)



\$120,17M

Total Sales

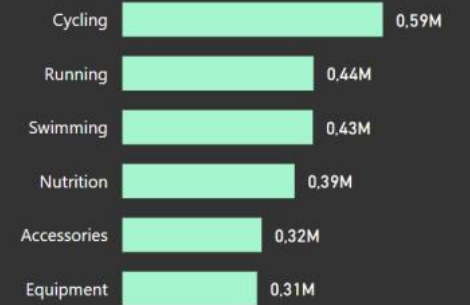


Sales



2M

Total Units Sold



Create visuals in Power BI Desktop



Power BI
Desktop

Create a matrix visual

1. Insert a matrix visual that displays [Revenue] in the "Values" well and [hr_cat] in "Lines".
2. Sort by [Revenue] descending. Using the ellipsis icon at the top right of the visual.
3. Expand and collapse [hr_cat] by using the arrows in the header icons or by right-clicking.
4. Apply conditional formatting (icons) to [Revenue].
5. Insert a segment on dimCategories[Category Name].
6. Add a title and logo.
7. Add a scrim in the background. Download the scrim library from github.com.
8. Rename the page 'Cat Hierarchy'.
9. Add a filter on all pages limiting the possible values of Calendar[Year] to 2020,2021 and 2022.

The 'Cat Hierarchy' page



Sales by categories, subcategories and products

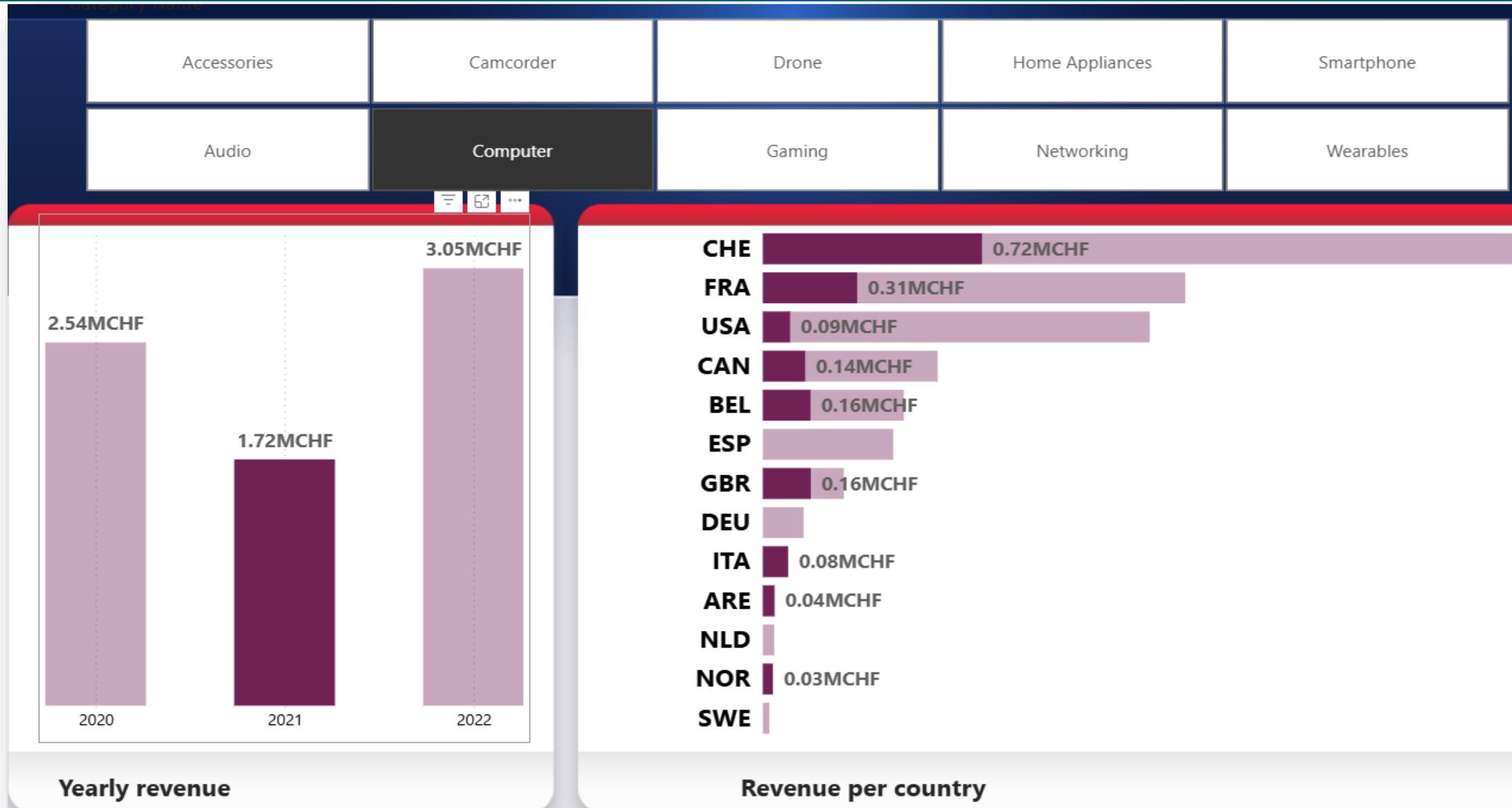
- ☐ Accessories
- ☐ Audio
- ☐ Camcorder
- ☐ Computer
- ☐ Drone
- ☐ Gaming
- ☐ Home Appliances
- ☐ Networking
- ☐ Smartphone
- ☐ Wearables

Cat	Revenue
⊕ Home Appliances	8'926'451 CHF
⊕ Accessories	7'715'752 CHF
⊕ Networking	7'431'413 CHF
⊕ Smartphone	7'362'838 CHF
⊖ Computer	7'311'023 CHF
Mini PC	● 2'319'817 CHF
Tablet	▲ 2'155'596 CHF
Laptop	▲ 1'567'650 CHF
Desktop	◆ 828'246 CHF
All-in-One	◆ 439'714 CHF
⊕ Camcorder	6'790'551 CHF
⊕ Wearables	6'297'579 CHF
⊕ Drone	6'066'240 CHF
⊕ Audio	5'741'527 CHF
⊕ Gaming	4'121'711 CHF
Total	67'765'085 CHF

Create a Columns + Bars visual

1. Insert a column visual with [Revenue] in "Y-Axis" and [Year] in "X-Axis".
2. Insert a segment on dimCategories[Category Name].
3. Add a horizontal bar visual with [Country] in "Y-Axis" and [Revenue] in "X-Axis".
4. Add the metrics [COGS], [Profit], and [NB sales] in the tooltips of the 2 visuals.
5. Add a scrim in the background and insert two titles below the visuals
6. Rename the page 'Revenue by Year/Country'.
7. Test Cross-Highlighting and Cross-filtering between the two visuals.

The page 'Revenue by Year / Country'



Create a Combo Visual

1. Create the 3 DAX measures that calculate the [Revenue] for 2020, 2021 and 2022:

Revenue 2020 = `CALCULATE([Revenue],Calendar[Year]=2020)`

2. Create the DAX measure that calculates the previous year's [Revenue].

Revenue LY = `CALCULATE([Revenue],SAMEPERIODLASTYEAR(Calendar[Date]))`

3. Insérer des visuels de type Carte affichant [Revenue], [Revenue LY], [Revenue 2020] , [Revenue 2021] et [Revenue 2022]
4. Insert a Combo visual with [Revenue] and [Revenue LY] in "Y-Axis of the column" and the year hierarchy in "X-Axis". Then add [NB Sales] in "Line Y-axis" and format the visual as shown on the next page.
5. Use the arrow icons to navigate through the different hierarchical levels of the visual.
6. Rename the page 'CY, LY and NB Sales'.



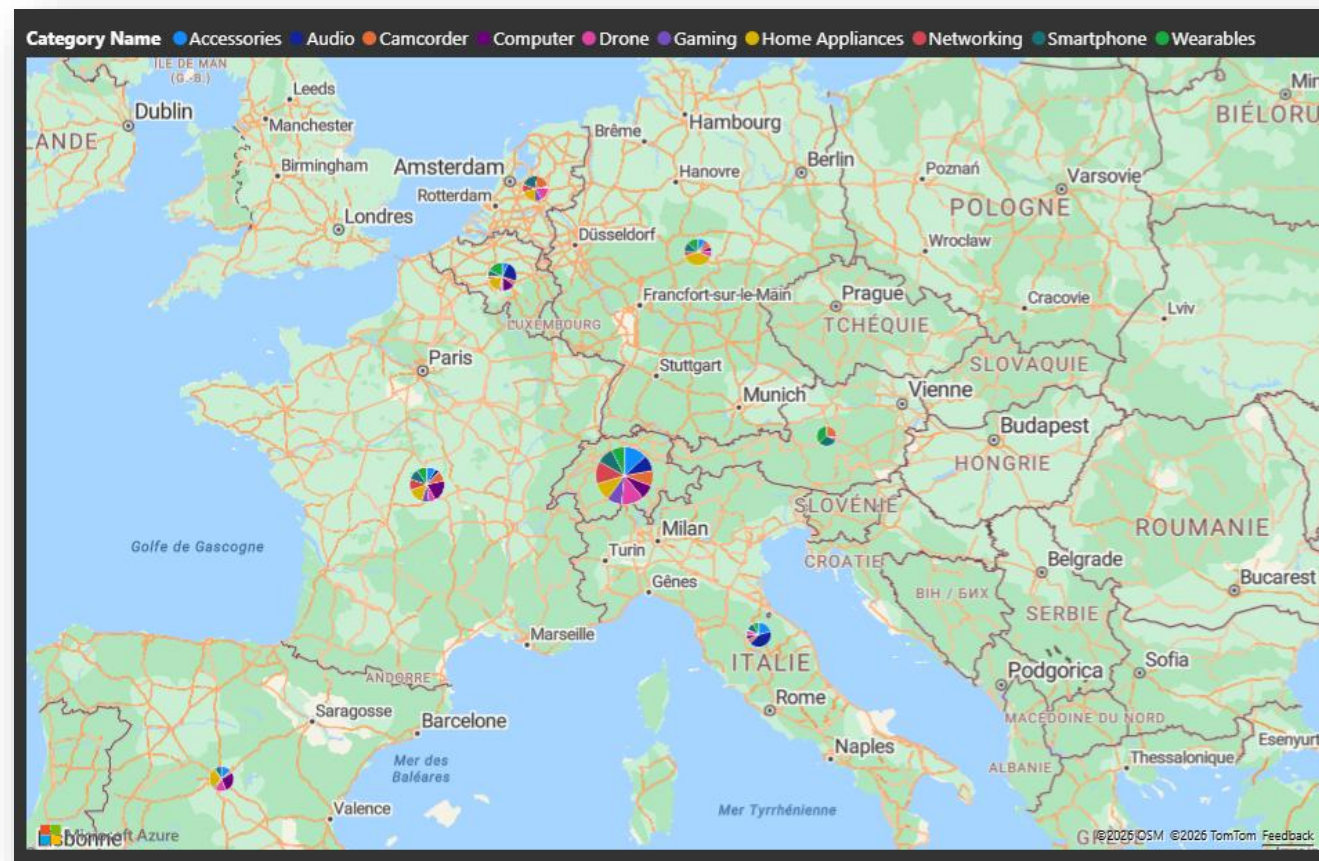
Create a visual with Small Multiples

1. Insert a clustered column visual with [Year] in "X-axis" and [Revenue] in "Y-axis".
 2. Drag [Category Name] into "Small Multiples".
 3. Add two reference lines (Format > Visual Object > Reference Lines), one for the maximum sales and the other for the average (shade the area below the line for the average).
- Display the data labels for the 2 rows with the Data label style to 'both'.



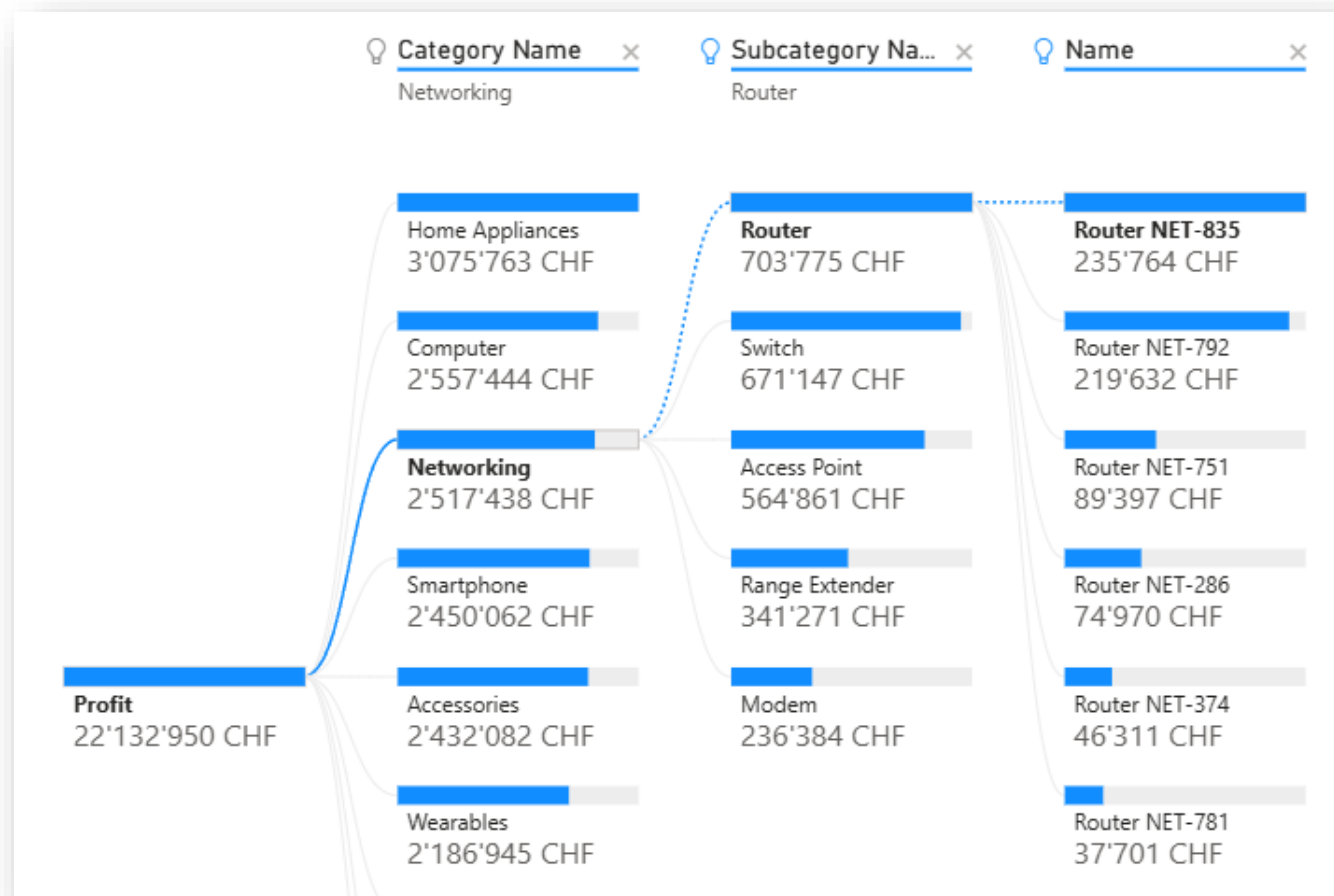
Create an azure map visual

1. Insert an 'Azure Maps' visual and drag `dimCountriesISO[English short name]` into "Location". Countries present in the model are now represented by bubbles on the map.
2. Add `[Revenue]` to the "Size" well. The size of the bubbles depends on the revenue value for that country.
3. Add `dimCategories[Category Name]` in "Legend" to replace the bubbles with pie charts showing the distribution of `[Revenue]` between the various categories.



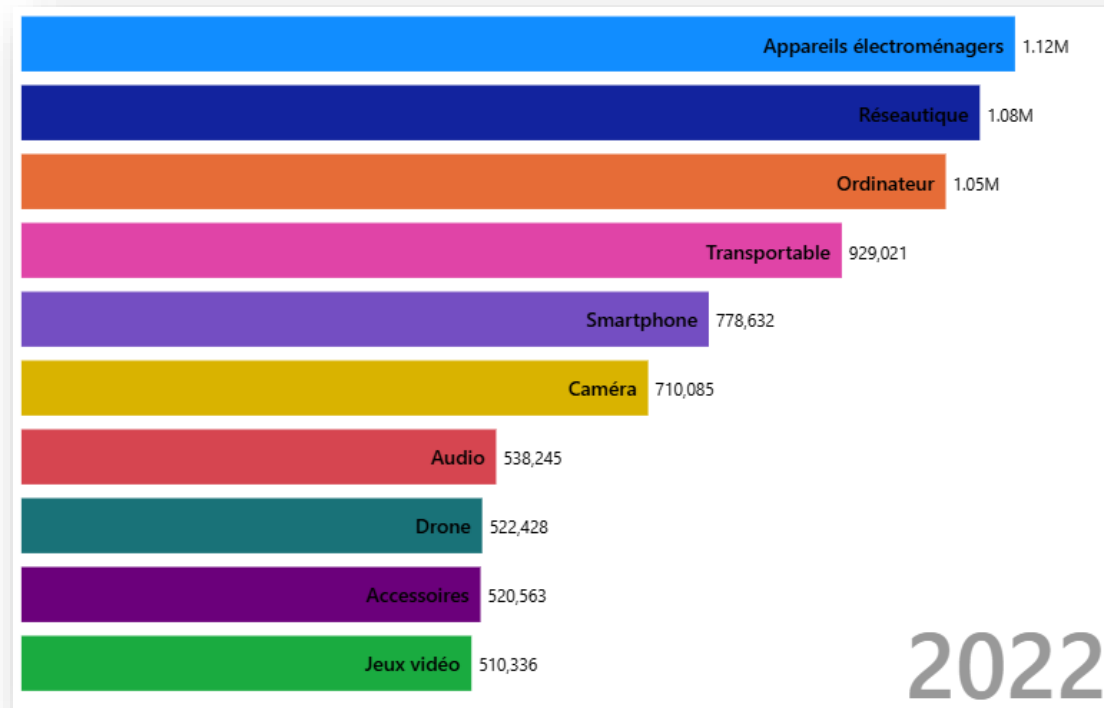
Analysis with a decomposition tree

- Insert a “decomposition tree” visual and drag FactCommands[Profit] into the “Analyze” well and dimCategories[Category Name], dimCategories[Subcategory Name] and dimCountriesISO[English short name] into 'Explain by'.



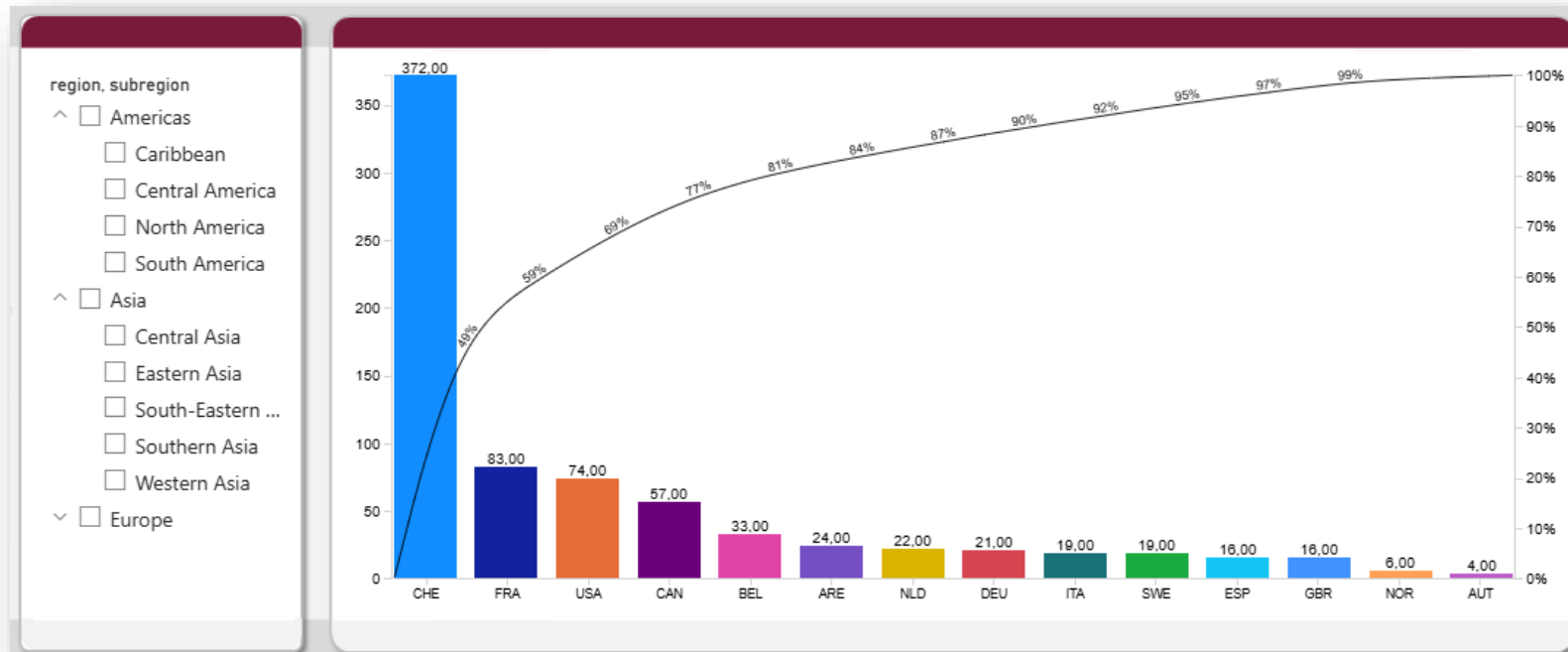
Insert an Animated Bar chart race visual

- Click on the 'Get more visuals' button and in the search bar, type 'animated bar chart race'. Choose the visual created by Wishyouulization and add it to Power BI.
- Insert this visual in a new page and drag Calendar[Year] into the "Period" well, FactOrders[Profit] into "Value" and dimProducts[Category Name] into "Name".
- Check the options in the Format panel, including "Duration" and "TopN"



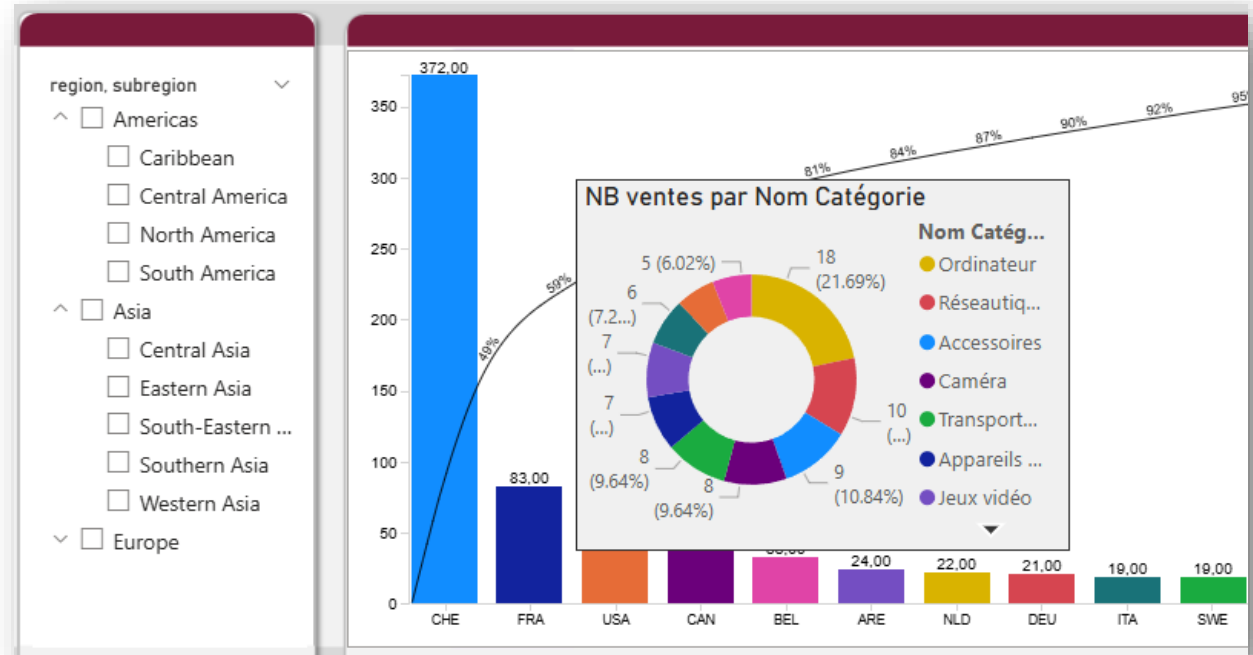
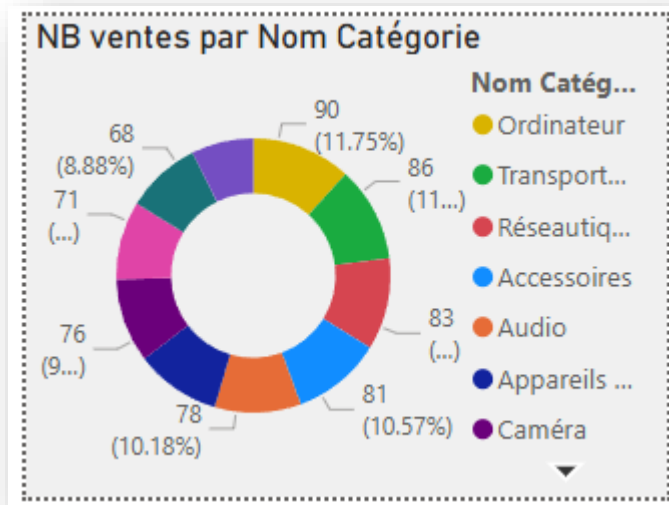
Insert a Pareto chart

- Click on the 'Get more visuals' button and in the search bar, type 'Pareto'. Choose the visual created by sio2Graphs and add it to Power BI.
- Insert this visual in a new page and drag dimCountriesISO[Alpha-3] into the "Categories" well and [NB Sales] into "Values".
- Insert a slicer and drag dimcountriesISO[region] and dimcountriesISO[subregion] into the "Field" well. Keep only the 'Europe', 'Asia' and 'North America' regions in the visual filter.
- Rename the page 'Pareto' and insert a scrim.



Insert a tooltip page

- Add a new page 'TT NB sales > cat'. In the page format, choose 'Tooltip' as the page type and drag the dimCountriesISO[Alpha-3] field into the 'Show tooltip on' well.
- Insert a donut visual with [NB Sales] in the "Values" well and dimCategories[Category Name] in "Legend". In Format > Size & Style > Background, choose a light color.
- Go back to the Pareto visual and check the tooltip window appearing on the country columns.



Use quick measures and visual calculation

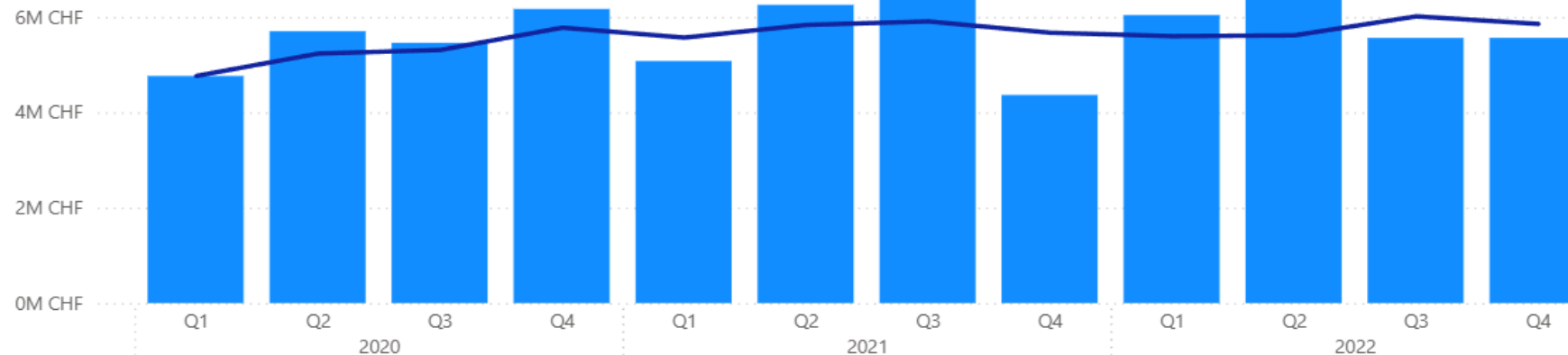
- In a new 'Quick/Visual' page, insert a table visual with Calendar[Year Quarter] in 'Rows' and [Revenue] in 'Values'.
- Click on Home > Quick Measure and select 'Year-to-date', drag [Revenue] into 'Base Value' and Calendar[Date] into 'Date'. A new DAX measure (quick measure) is created, add it to the table visual.
- Create another quick measure by choosing 'Cumulative Result', drag [Revenue] into 'Base Value' and Calendar[Date] into 'Date'. Add the new measure to the table visual.
- Click Home > New Visual Calculation. Click on the fx button and select 'Parent Percentage'. In the formula, replace [Field] with [Revenue] and [Axis] with [Quarter Year]. A visual calculation is added to the table. To modify the latter, you have to click the pencil icon to the right of the name in the "Values" well.
- In order to display the percentage format, modify the formula for the visual calculation as follows:
Parent Percentage = `FORMAT(DIVIDE([Revenue], COLLAPSE([Revenue], [Year Quarter])), "0.0%")`
- Insert a Combo visual and drag Calendar[Year Quarter] into "X-Axis", [Revenue] into "Y-Axis of Column". Sort ascending on [Year Quarter].
- Add a new 'Moving Average' visual calculation. Replace [Field] with [Revenue] and Window size with 3 to average over 3 periods. Once the visual calculation is created, drag it into "Line Y-axis."

Quick Measures and visual calculations

Year/Quarter	Revenue	Revenue %	Running total YTD	Running total on REVENUE
2020/Q1	4'762'756 CHF	7.0%	4'762'756 CHF	4'762'756 CHF
2020/Q2	5'701'358 CHF	8.4%	10'464'114 CHF	10'464'114 CHF
2020/Q3	5'455'708 CHF	8.1%	15'919'822 CHF	15'919'822 CHF
2020/Q4	6'167'397 CHF	9.1%	22'087'219 CHF	22'087'219 CHF
2021/Q1	5'075'765 CHF	7.5%	5'075'765 CHF	27'162'984 CHF
2021/Q2	6'253'177 CHF	9.2%	11'328'942 CHF	33'416'161 CHF
2021/Q3	6'388'147 CHF	9.4%	17'717'089 CHF	39'804'308 CHF
2021/Q4	4'363'558 CHF	6.4%	22'080'647 CHF	44'167'866 CHF
2022/Q1	6'038'581 CHF	8.9%	6'038'581 CHF	50'206'447 CHF
2022/Q2	6'437'539 CHF	9.5%	12'476'120 CHF	56'643'986 CHF
2022/Q3	5'559'780 CHF	8.2%	18'035'900 CHF	62'203'766 CHF
Total	67'765'085 CHF	100.0%	23'597'219 CHF	67'765'085 CHF

Revenue and moving average by Year, Quarter and Month

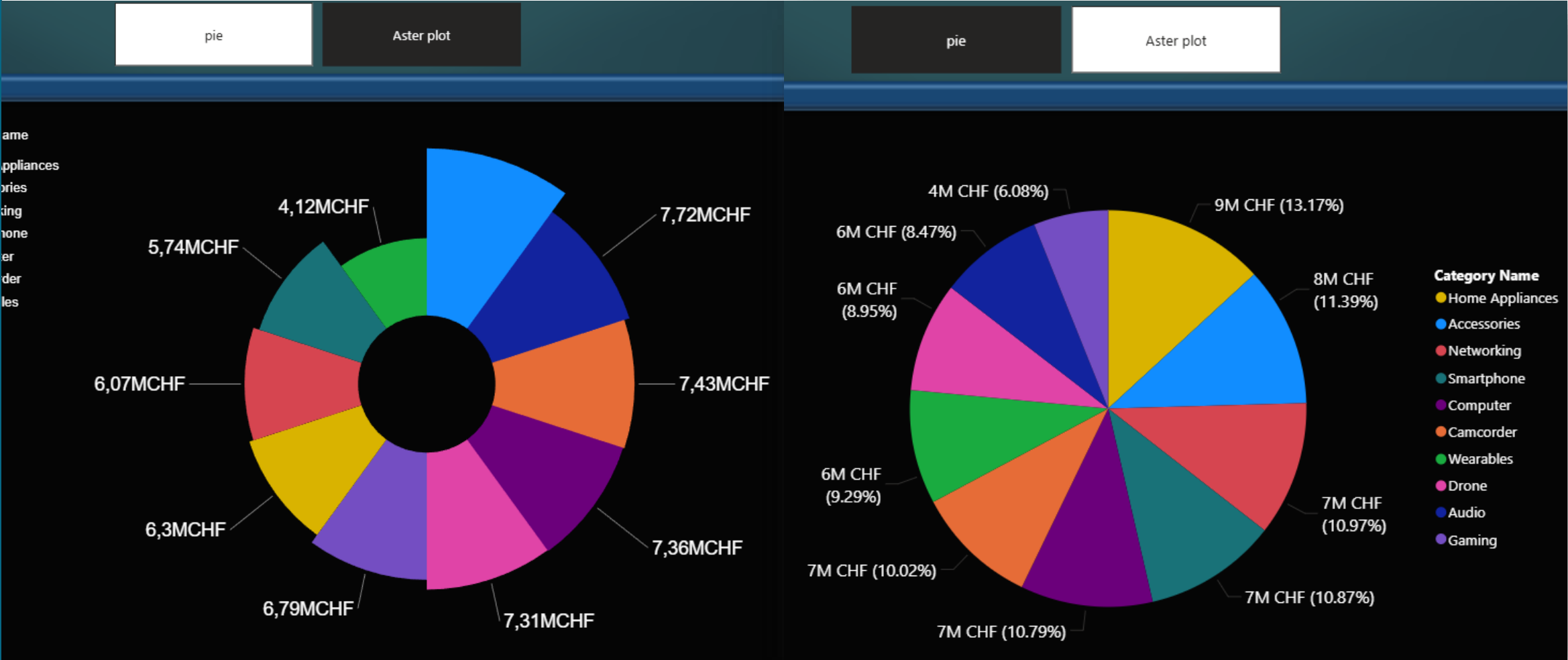
● Revenue ● Moyenne mobile



Insert two visuals into one page

- Add a new 'Pie and Aster by category' page. Add a Pie chart visual with [Revenue] in the "Values" well and dimCategories[Category Name] in "Legend".
- Download a new visual, search for 'Aster plot' from Microsoft and add it to Power BI. Drag [Revenue] into the "Y-Axis" well and dimCategories[Category Name] into "Category".
- Display the Bookmarks pane (View > Bookmarks). Open the selection pane (Show > Selection) and hide the aster plot visual. Center the pie chart visual and save the bookmark in this state (Bookmarks pane> Add). Rename it 'Pie'.
- In the selection pane, hide the pie chart visual and show the Aster plot visual. Add a new bookmark in this page state. Rename it 'Aster plot'.
- In the Bookmarks pane, right-click on one of the bookmarks and select 'Group'. Drag the 2nd bookmark into the newly created group. Rename the group 'Aster-pie'.
- On the report page, Insert > Buttons > Navigator > Bookmark Navigator.
- Select the bookmark Navigator and go to Format > Bookmarks and choose the 'Aster-pie' group created earlier from the list of available bookmarks.
- In Power BI Desktop, you need to press Ctrl+click to click on any button such as the bookmark navigator buttons.

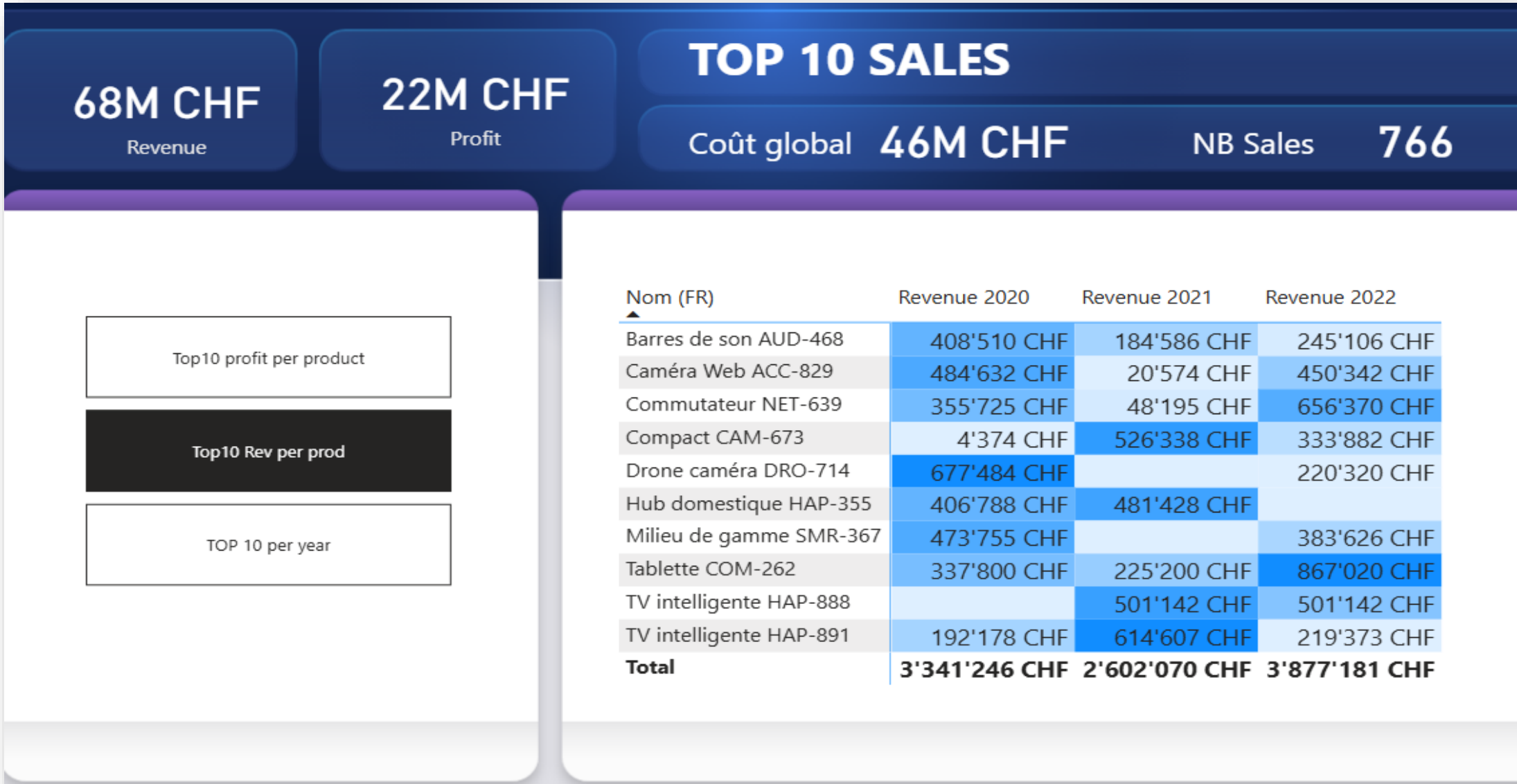
Alternate between two visuals with bookmarks



Insert three Top10 visuals with bookmarks

- Add a new 'TOP10' page. Add a funnel visual with [Profit] in the "Values" well and dimProducts[Name (EN)] in "Category". In the Filters pane, go to Filters on this visual and choose 'Top N' filter type for [Name (EN)] and then select 'Top' and '10' in 'Show Items'. Then drag [Profit] into "By Value". Click on Apply Filter. The funnel is now limited to the 10 highest profits per product name.
- Insert a matrix visual with dimProducts[Name (EN)] in "Lines" and [Revenue 2020], [Revenue 2021] and [Revenue 2022] in the "Values" well.
- Insert another matrix with dimProducts[Name (EN)] in "Lines" and [Revenue], [Profit] and [NB Sales] in the "Values" well. Apply conditional formatting to the [Profit] column. Sort on [Revenue] descending.
- Display the Bookmarks pane (View > Bookmarks). Open the selection pane (Show > Selection) and hide two out of three visuals and save a bookmark in that state (Bookmarks > Add). Give it an appropriate name.
- Repeat this for the other two visuals to get three separate bookmarks, one for each visual.
- In the Bookmarks pane, right-click on one of the bookmarks and select 'Group'. Drag the other 2 bookmarks into the created group. Rename the group 'TOP 10'.
- On the report page, Insert > Buttons > Navigator > bookmark navigator. In Format > Grid Layout > Orientation, choose 'Vertical' and place the buttons to the left of the visuals. In Format > Bookmarks > Bookmark, choose the 'TOP 10' group from the list of available bookmarks.
- Insert card visuals with the following metrics: [Revenue], [Profit], [COGS], and [NB Sales]. Insert headings to finalize the presentation.

Page with bookmark navigator TOP 10



Create a parameter table

1. Modeling > New Parameter > Fields

Select the measures in the measures table (they then appear in the list on the left)

Check the option 'Add slicer...' if you haven't already.

2. A table will be added to the model.

3. A slicer is also added to the page with the names of the measures in the buttons. Change the slicer display type if necessary.

Parameters

Add parameters to visuals and DAX expressions so people can use slicers to adjust the inputs and different outcomes. [Learn more](#)

What will your variable adjust?

Fields

Name

Parameter

Add and reorder fields

Revenue

×

COGS

×

Profit

×

Margin %

×

Revenue LY

×

NB Sales

×

☒ Add slicer to this page

Fields

Search

01 MEASURES

☐ AVG Ventes

☒ COGS

☐ DT title

☒ Margin %

☒ NB Sales

☒ Profit

☒ Revenue

☐ Revenue 2020

☐ Revenue 2021

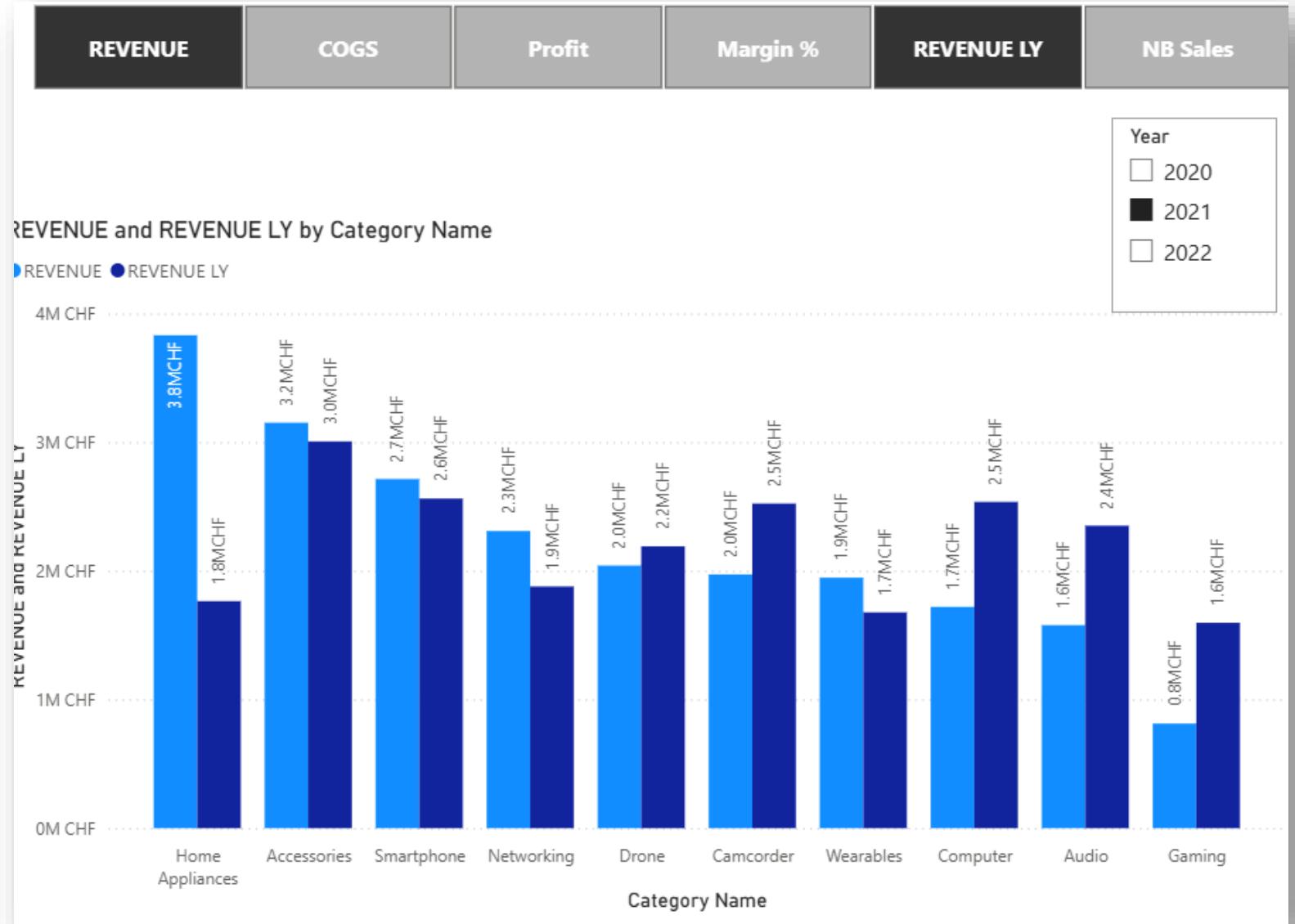
☐ Revenue 2022

☒ Revenue LY

Create

Insert a multi-measures visual

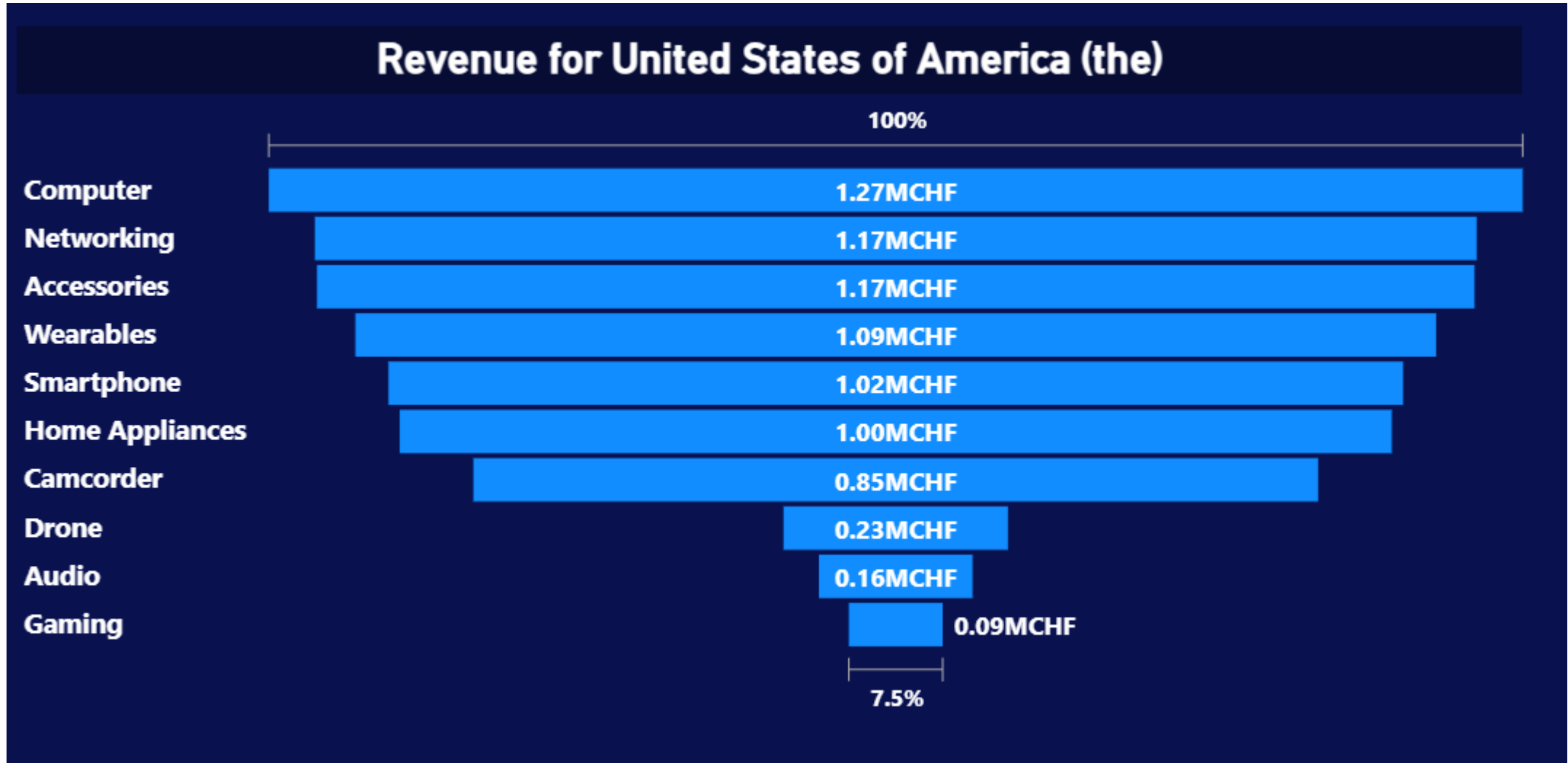
1. Insert a visual in clustered columns with the 'Parameter' field of the table of the same name in the Y axis and the name of the category in the X axis.
2. To compare multiple measures with each other, Ctrl + click on the desired measures in the slicer to display them to the visual.
3. Add a slicer with Calendar[Year] in "Field".



Drillthrough Page over Country

- Create a new page with a funnel visual with [Revenue] in "Values" and dimProducts[Cat] in "Category".
- Go to the Page Format pane and insert:
 - ◇ Information on the page > Name: 'DT Alpha3'.
 - ◇ Page Information > Page Type: 'Drillthrough'
 - ◇ Information on the page > Drill through from > dimCountriesISO[Alpha-3]
 - ◇ Canvas Background > Color: Choose a dark blue.
- Create a DAX measure displaying a dynamic headline:
DT title = "Revenue for country "& **SELECTEDVALUE**(dimCountriesISO[English short name])
- Insert a text box and go to Format > Title > Title > fx and select the DAX measure [DT title] from the drop-down list at the bottom of the window.
- The drillthrough page is now active on any visual displaying [Alpha-3].

The 'DT Alpha3' drillthrough page

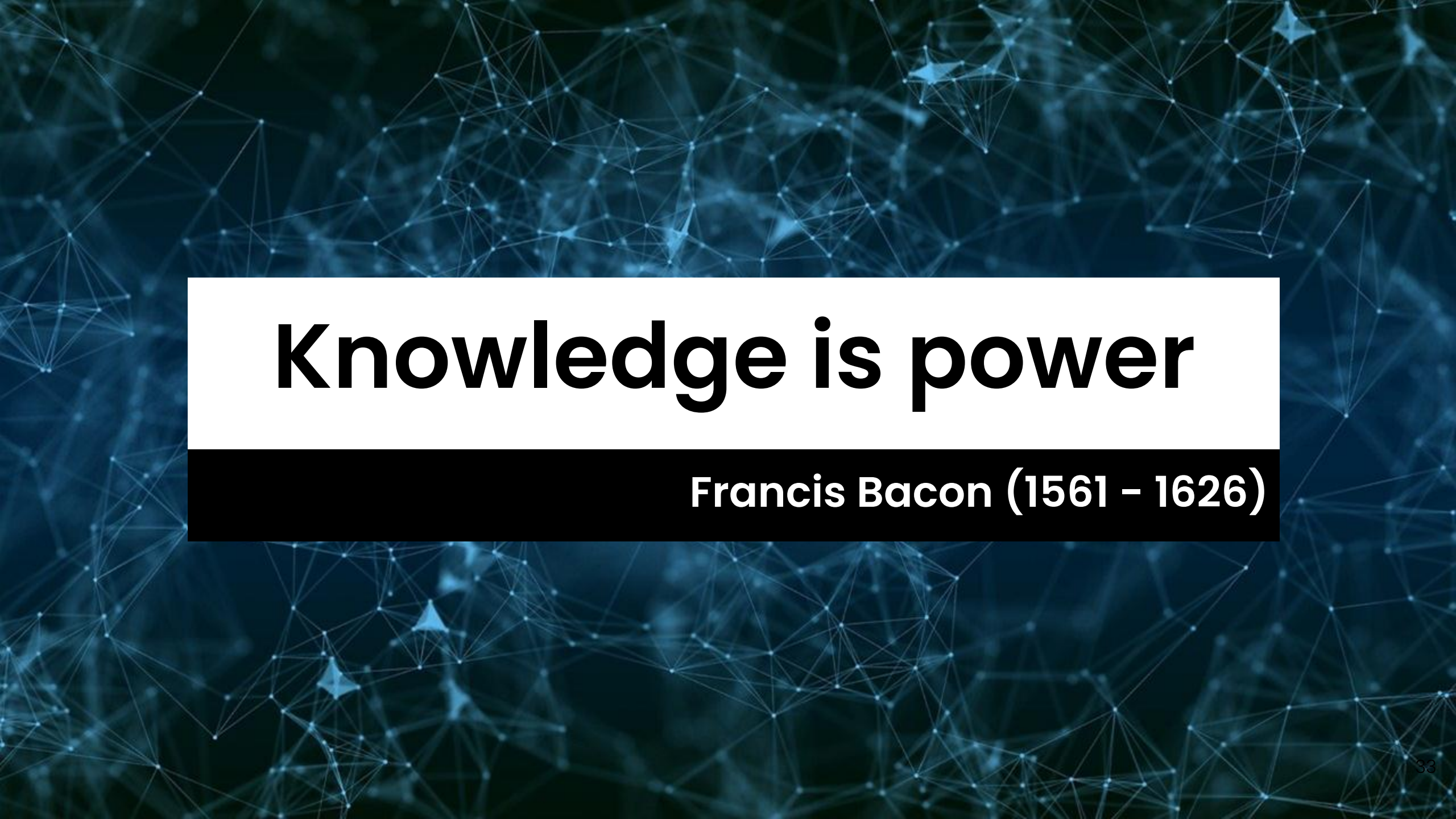


Revenue drillthrough page

- Create a new page and add a table visual. Drag the following fields into the “Columns” well:
 - ◇ dimClients[Client Name] and dimClients[City]
 - ◇ [Revenue], [Profit], [Margin %] and [NB Sales]
- Go to the Page Format pane and insert:
 - ◇ Page Information > Name: 'DT Revenue'.
 - ◇ Page Information > Page Type: 'Drillthrough'
 - ◇ Page Information > Drill through from > [Revenue]
 - ◇ Canvas Background > Color > (Choose a color).
- Insert a text slicer with dimClients[Client Name] in “Field”.
- Insert a slicer on Calendar[Year].
- This drillthrough page is now active on any visual displaying [Revenue].

Publish the report to Power BI service

- When the report is ready, click Home > Publish to publish the report and semantic model to the cloud in one of your workspaces (by default 'my workspace').
- Open the report in the Power BI service and select visuals to pin to a new dashboard (to be created when prompted).
- Try pinning an entire page to an existing dashboard.
- Add tiles to your dashboard, i.e. the company logo.
- The dashboard can now be shared with your colleagues (if they are using a PRO licence or above).



Knowledge is power

Francis Bacon (1561 – 1626)